

Artificial Intelligence

BCA — Semester 4 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What is knowledge, representation and reasoning? When a machine is said to be passed Turing test? Give any two examples of constraint satisfaction problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Why CNF is necessary? From the following facts, show that "Charlie is a mammal" using FOPL based resolution method. • Cows, pigs and horses are mammals • The child of a horse is a horse • Bluebeard is a horse • Bluebeard is Charlie's father • Child and father are inverse relations • Every mammal has a father.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Justify that AI can't exist without searching? Solve the following 8 puzzle problem using Hill climbing search algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain the different ambiguities related to NLP.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Differentiate between goal based agent and utility based agent.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Describe the concept of learning by analogy with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is the task of activation function? What are its types?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How Dempster - Shafer theory can be used for reasoning with certainty? Explain with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define expert system. Describe its architecture.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is the purpose of alpha beta pruning? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Illustrate your own scenario for shopping at my mall and represent those concepts using scripts.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Define deterministic and non-deterministic environment. Differentiate between BFS and DFS.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.